

IT58 - FINAL DELIVERABLE

Human-Centered AI Project

Pest and Disease Advisory Module: AI Decision Support for Rice Farmers

Group Name: Rice Farmers AI Advisory Group

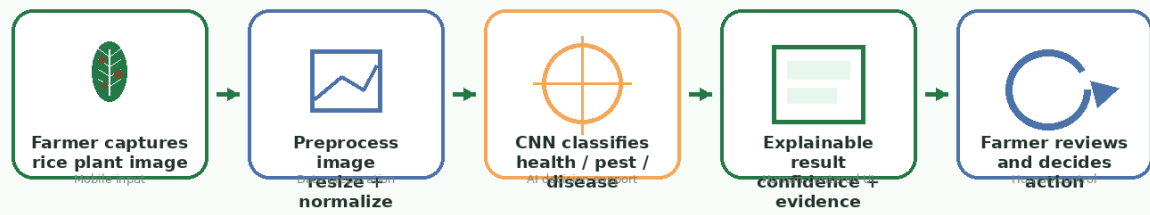
Course: IT 58 Emerging Technologies in HCI

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Target Community	Rice farmers, agricultural technicians, and farm cooperatives in rural farming communities
Project Type	Decision Support / Advisory / Assistive AI
Document Type	Single PDF final deliverable with embedded screenshots and prototype section

Concept Sketch / Idea Diagram

AI advisory support from rice-plant photo to farmer-controlled action



Core design rule: the AI suggests and explains. The farmer remains the final decision-maker.

Figure 1. Concept sketch showing the farmer-controlled AI advisory flow.

Document basis: This final deliverable consolidates the uploaded project documentation, the AI design metaphors activity, and the official IT58 final deliverable instructions.

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Important submission note: A working Figma or deployed prototype URL was not included in the uploaded files. This PDF includes a clickable placeholder and complete embedded prototype screens. Replace the placeholder link with the actual group prototype link before final submission.

1. Project Overview

Project Title

Pest and Disease Advisory Module: AI Decision Support for Rice Farmers

Project Alignment

The project is a human-centered AI advisory system that helps rice farmers recognize possible pests and plant diseases at an early stage through image recognition. The AI output is not designed to replace the farmer. It supports the farmer by identifying likely crop-health problems, explaining the basis of the result, and recommending treatment or prevention steps while preserving user control.

Target Community

The target community is rural rice-farming communities, especially small-scale rice farmers with limited access to agricultural experts. Secondary users include agricultural technicians, extension workers, farm cooperatives, and agriculture organizations that support rice production.

Problem Statement

Rice farms are vulnerable to pests and diseases that can spread quickly and reduce harvest yield. Many farmers in rural areas cannot immediately consult agricultural experts, so crop-health issues may be identified late or treated incorrectly. The project addresses this gap by providing a mobile AI advisory tool that analyzes rice-plant images and returns understandable, actionable recommendations.

Objectives

Objective	Description
1. Early detection	Allow farmers to capture rice-plant images and receive a quick advisory result for likely pest, disease, or healthy condition.
2. Explainable advice	Show confidence score, visible evidence, alternative outputs, and plain-language explanation so users understand the result.
3. Farmer control	Provide accept, reject, retake, report error, and consult technician options before any real-world action is taken.
4. Responsible data use	Collect only necessary data, request consent, and allow optional image use for future model improvement.
5. Continuous improvement	Use feedback, correction reports, and outcome records to improve the dataset, model performance, and recommendations over time.



Figure 2. Original project workflow screenshot from the uploaded HCI project documentation.

2. Community Need Analysis

Description of Users

The primary users are rice farmers who need fast, clear, and low-cost crop-health support while working in the field. These users may have limited technical background, inconsistent internet connectivity, and limited access to specialists. Agricultural technicians and cooperatives are supporting users because they can verify unclear cases, help interpret advisory results, and guide proper treatment.



Figure 3. Persona screenshot showing primary, support, and community users.

Identified Needs and Problems

Need / Problem	Design Response
Delayed diagnosis of pests and diseases	Provide quick mobile image analysis and immediate advisory output.
Limited expert access in rural areas	Add consult-technician flow for cases requiring human expert review.
Risk of wrong treatment	Show confidence, alternatives, warnings, and override controls before action.
Low technical literacy	Use simple terms, icons, readable text, and step-by-step recommendations.
Unclear or poor-quality images	Add camera guide, image quality check, and retake-photo prompt.
Privacy concern over uploaded plant images	Use consent screen, optional data sharing, anonymization, and deletion policy.

Data Sources

The project requires images of healthy rice plants, rice plants affected by pests, rice plants affected by diseases, and comparative examples under varied field conditions. It also requires validated treatment and prevention information. Potential sources include agricultural research institutions, public agricultural datasets, government agriculture departments, universities, and consented farm image collection.

Methods Used

The analysis used document review, persona development, problem mapping, task workflow analysis, AI system decomposition, failure-mode analysis, and prototype heuristic evaluation. These methods are appropriate because the project focuses on a practical human-AI interaction where user trust, safety, accessibility, and control must be considered together.

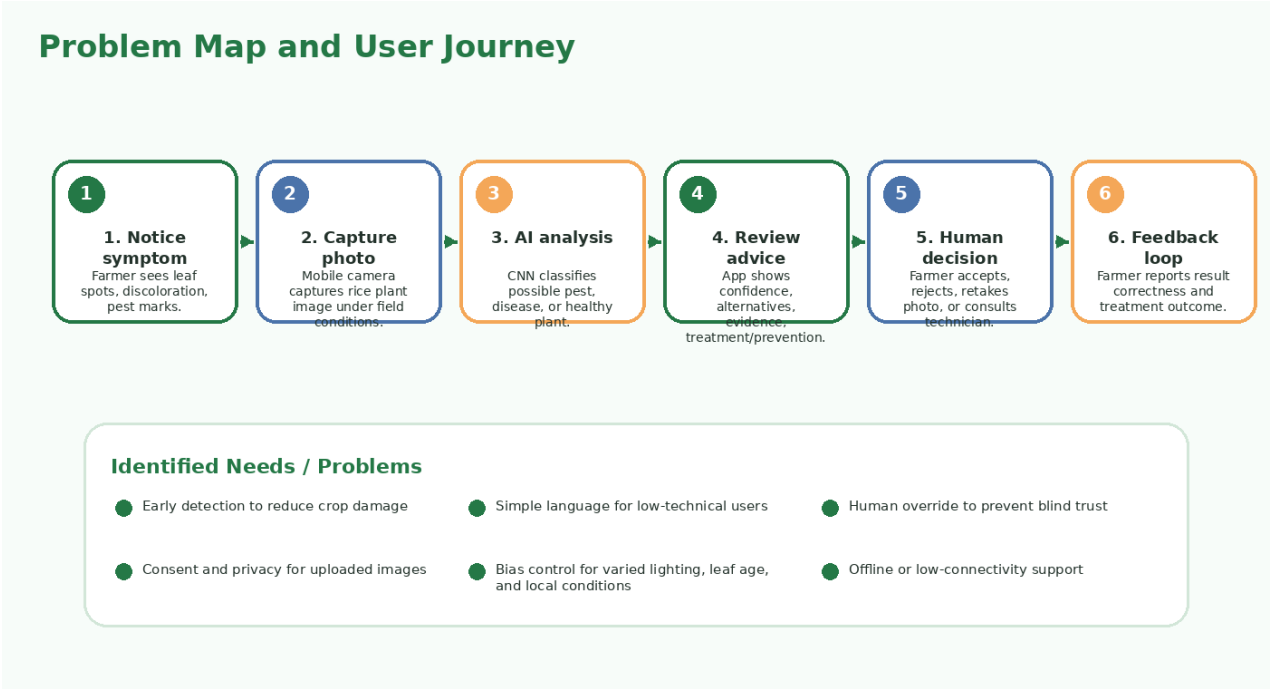


Figure 4. Problem map and user journey screenshot.

3. AI System Design

How AI Is Used

The AI component is used as an image classification model. A farmer captures a photo of a rice plant, the system preprocesses the image, and a CNN-based model classifies the plant condition as a pest type, disease type, or healthy plant. The predicted result is then mapped to treatment, prevention, and best-practice recommendations.

Input -> Process -> Output

Stage	Details
Input	Rice plant image captured from a mobile device. Optional context may include growth stage, local condition, severity note, and date/time.
Process	Image resize, normalization, clarity check, CNN classification, confidence scoring, top alternatives, recommendation mapping, and warning logic.
Output	Predicted condition, confidence score, alternative outputs, highlighted visual evidence, explanation text, treatment/prevention advice, and user-control options.

Example Input and Output

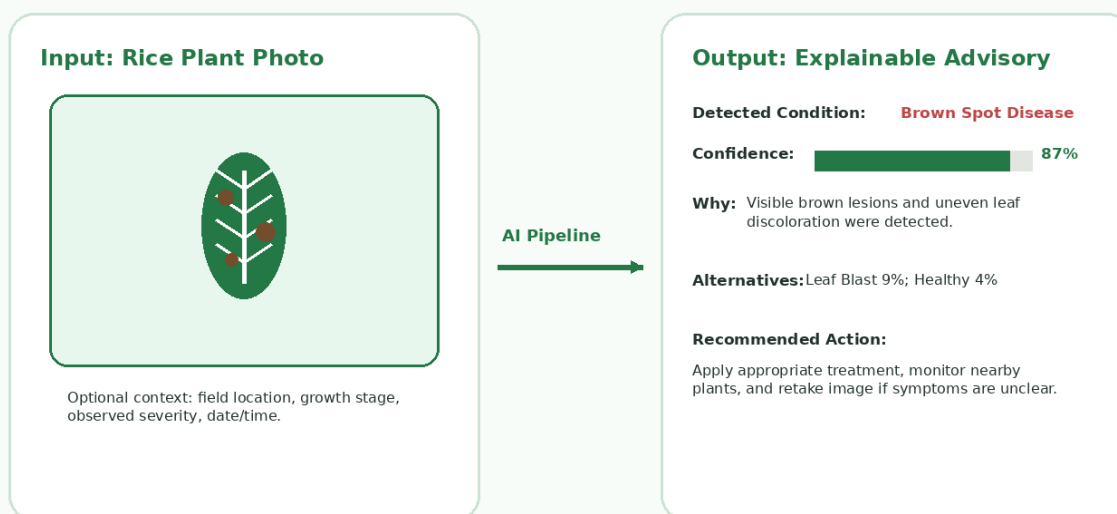


Figure 5. Example input and explainable output screenshot.

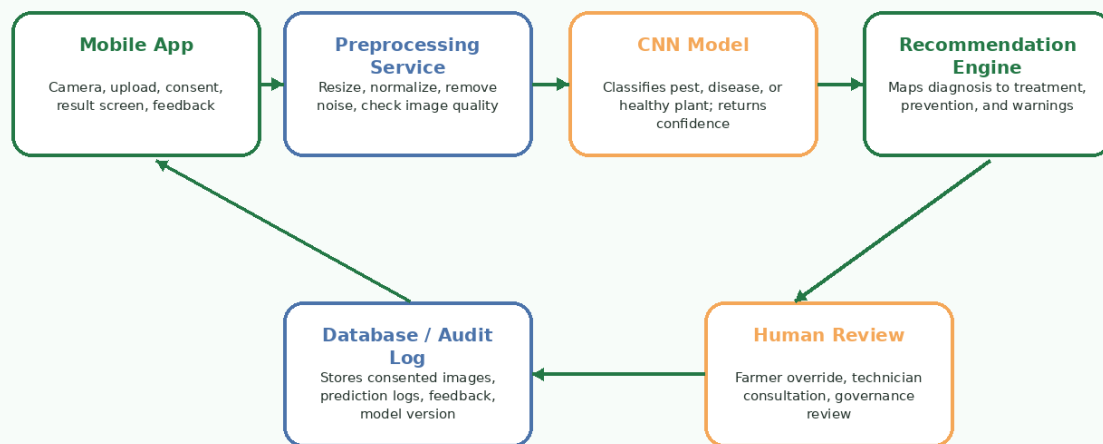
AI Approach

The recommended AI approach is a Convolutional Neural Network (CNN) for rice-plant image classification. A CNN is appropriate because pests and plant diseases often appear as visual patterns such as discoloration, lesions, leaf texture changes, and damaged areas. The system should return top-3 results, not only a single label, so that uncertain or visually similar cases can be reviewed by the farmer or a technician.

Data Plan

Data Component	Plan
Image classes	Healthy rice plant, common rice diseases, and common rice pest damage classes.
Labeling	Labels should be verified by agricultural technicians or reliable agricultural references.
Diversity	Collect varied lighting, background, angle, growth stage, severity, and local field conditions to reduce bias.
Quality control	Remove unusable images, check duplicate images, and flag unclear cases for expert review.
Training and testing	Use separated training, validation, and test sets. Test against unseen images before pilot deployment.
Feedback loop	Use farmer corrections, technician review, and treatment outcomes for future model evaluation and dataset refinement.

AI System Architecture



Safety layer: low confidence results trigger retake-photo, alternative outputs, or consult-technician flow.

Figure 6. System architecture diagram screenshot.

4. Human-Centered Features

This section addresses all required human-centered AI features: explainability, human override and appeal, privacy and consent, bias and risk analysis, accessibility, and governance/audit.

Explainability

The diagnosis screen should show what the AI detected, how confident it is, and why the result was suggested. The interface should include the predicted condition, confidence percentage, top alternative predictions, highlighted image region, and short explanation text. This avoids blind trust and helps the farmer decide whether the advisory result is reliable enough to use.

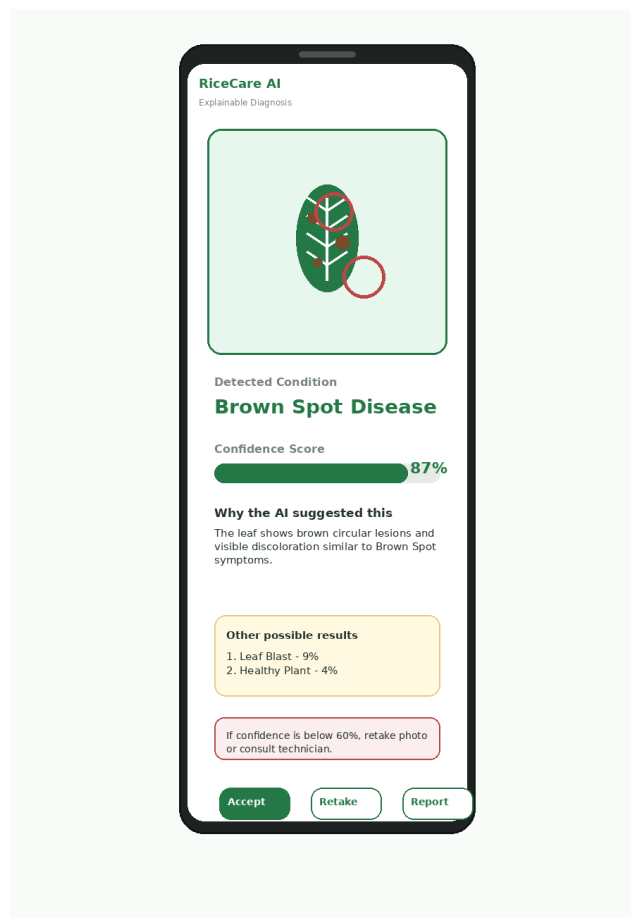


Figure 7. Explainable output interface screenshot.

Human Override and Appeal

The system must allow farmers to accept, reject, retake the photo, view alternatives, report an incorrect diagnosis, and consult an agricultural technician. This keeps the system at Level 1 autonomy: the AI suggests only, while the farmer remains the final decision-maker. The appeal mechanism is the report-error and consult-technician workflow.

Privacy and Consent

Before any photo is uploaded or stored, the app must show a clear consent screen. Required consent should be limited to image analysis. Optional consent should be separate for saving farm

history, using anonymized images for model improvement, or sharing with a technician. The system should minimize data collection, protect user records, and offer deletion where feasible.

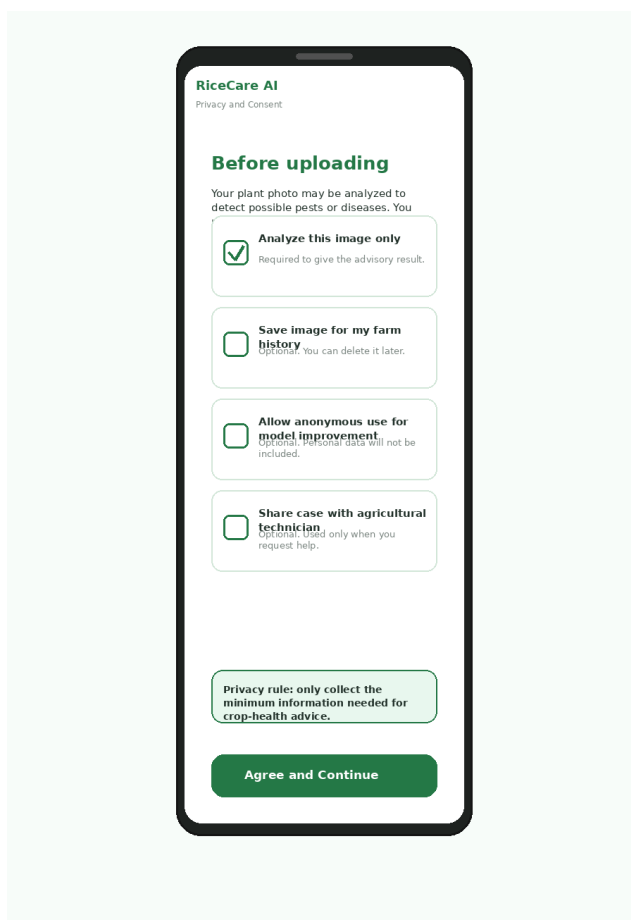


Figure 8. Consent and privacy UI screenshot.

Bias and Risk Analysis

Risk / Failure Mode	Impact	Mitigation
Wrong pest or disease prediction	Farmer may apply the wrong treatment or delay proper intervention.	Use expert-labeled datasets, top-3 alternatives, warnings, and technician review.
Healthy plant classified as diseased	Unnecessary cost, labor, and chemical use.	Show confidence and ask for verification before treatment.
Serious disease classified as healthy	Disease may spread and cause crop loss.	Set conservative thresholds; low-confidence cases should recommend retake or expert consultation.
Unseen pest/disease class	Model may force a wrong known label.	Add unknown/uncertain status and feedback collection for new cases.
Poor image quality	Incorrect or unusable result.	Use camera guide, quality check, retake-photo option, and no-result handling.
Over-reliance on AI	User may treat advice as certain.	Use clear warnings: AI is advisory, farmer/technician verification remains important.

Accessibility

The interface should use large readable text, high contrast, simple icons, clear buttons, local-language support where possible, and read-aloud options. Since farmers may use the app outdoors, the app should be readable in sunlight and should save drafts when connectivity is weak.

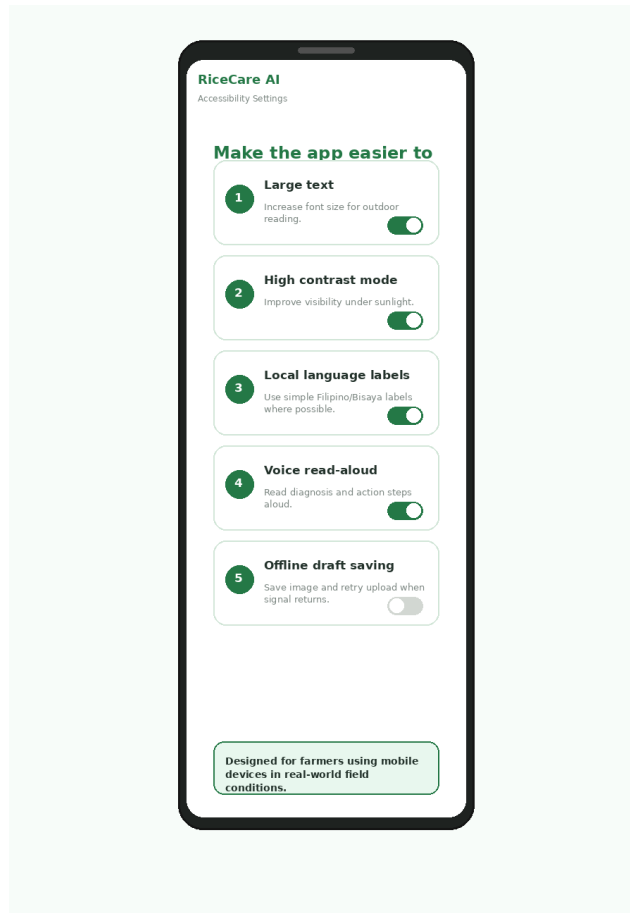


Figure 9. Accessibility feature screenshot.

Governance and Audit

The system should log the prediction ID, model version, input quality status, confidence score, consent status, farmer action, and feedback outcome. These logs support troubleshooting, model improvement, and accountability. Governance review should prioritize severe false negatives, repeated error reports, and data quality issues.

5. UI Prototype

Interface Description

The prototype is a mobile-first advisory application designed for field use. The design uses a clean green-and-white visual system associated with agriculture, large buttons, simple labels, and a short step-by-step flow. The main user journey starts with consent, proceeds to image capture and analysis, and ends with explainable diagnosis, advisory recommendations, and feedback.

Prototype Link: <https://figma.com/proto/REPLACE-WITH-ACTUAL-PROTOTYPE-LINK>

Replace the placeholder above with the actual Figma or deployed prototype link before submission. The UI screens are embedded below so the PDF remains self-contained.

Key Workflows

Workflow	Steps
New diagnosis	Welcome -> Consent -> Home -> Camera Guide -> Review Image -> Analyzing -> Diagnosis -> Advice
Low confidence result	Diagnosis -> Warning -> Retake Photo or Consult Technician
User override	Diagnosis -> Reject Result -> Report Error or View Alternatives
Feedback loop	Diagnosis/Advice -> Feedback -> Correct/Incorrect -> Treatment Outcome
Privacy control	Consent -> Required analysis only or optional saving/model-improvement consent

UI Prototype Screens

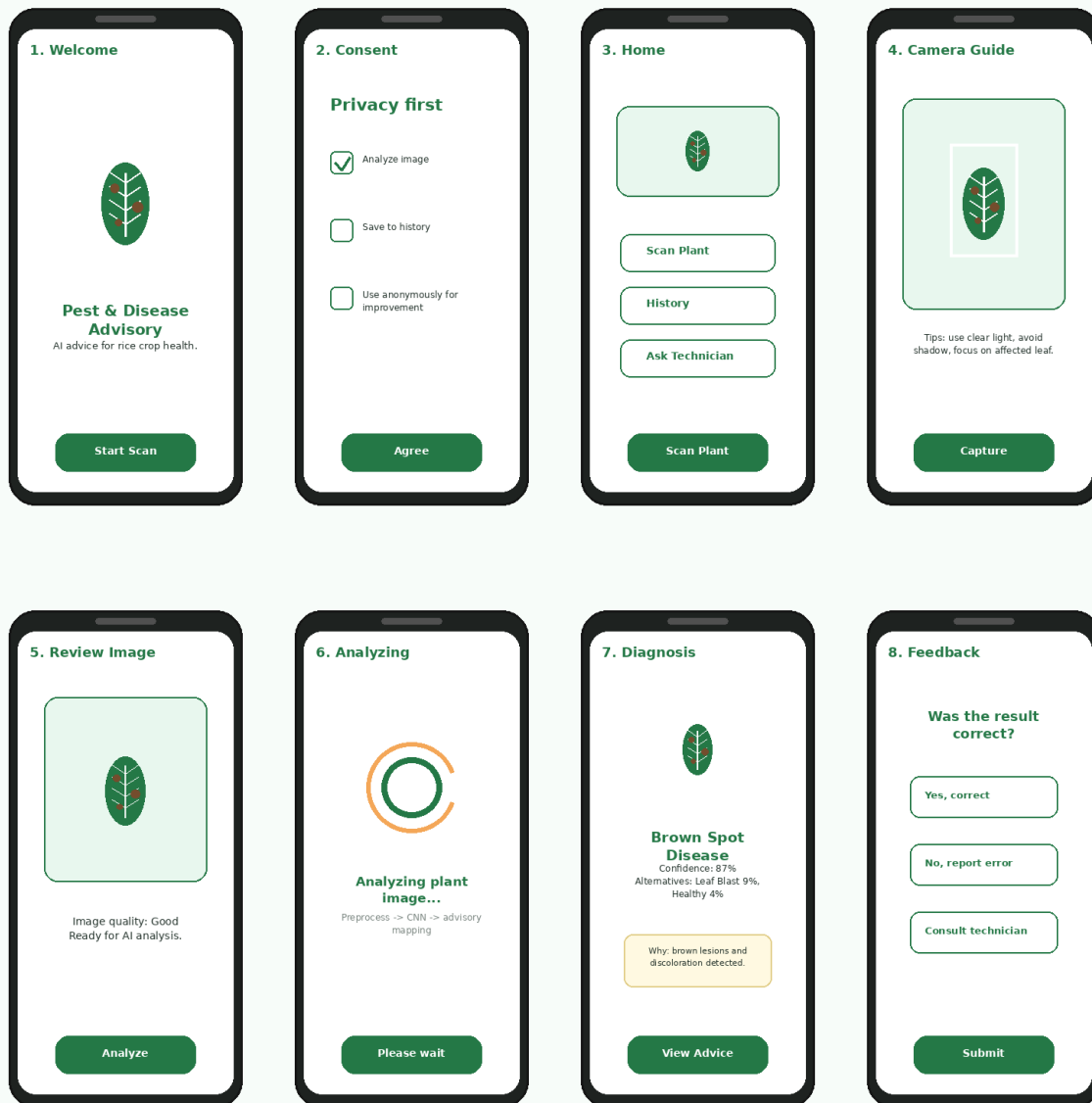
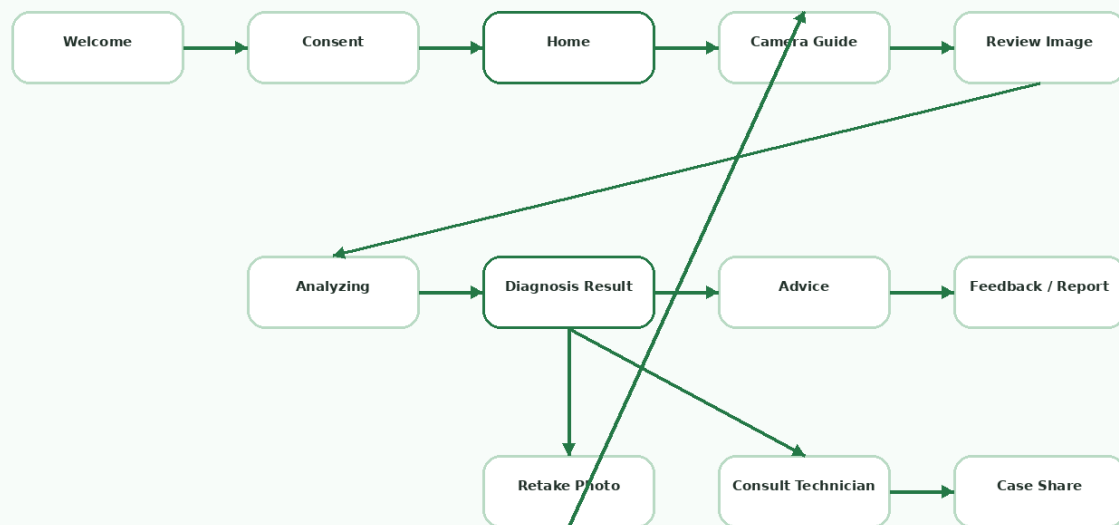


Figure 10. Eight embedded UI prototype screens.

Navigation Flow



Main path: consent -> scan -> analyze -> explain -> advice -> feedback. **Safety path:** retake, report error, or consult technician.

Figure 11. Navigation flow screenshot.

6. HCAI Evaluation

Evaluation Criteria

The prototype was evaluated using human-centered AI criteria: transparency, user control, safety, privacy, accessibility, fairness/bias control, reliability, and accountability. Because no live farmer testing evidence was included in the uploaded files, the current evaluation is a prototype-level heuristic review and scenario-based walkthrough. The recommended next step is field pilot testing with actual rice farmers and agricultural technicians.

Evaluation Evidence: Heuristic Test Summary

Criterion	Test Question	Finding	Status
Transparency	Can user understand why result was suggested?	Confidence, top alternatives, and highlighted region are shown.	Pass
Human Control	Can user reject, retake, or report result?	Override controls are visible on diagnosis screen.	Pass
Privacy	Is consent asked before upload/training use?	Consent screen separates required and optional data use.	Pass
Safety	What happens on low confidence?	Warning recommends retake or technician consultation.	Pass
Accessibility	Can low-technical users read and navigate?	Large text, high contrast, icon-based navigation, read-aloud option.	Pass
Bias Risk	Are varied field conditions considered?	Data plan includes varied lighting, angles, stages, and local classes.	Needs pilot

Evaluation type: prototype heuristic review and scenario-based walkthrough. Recommended next step: pilot test with actual rice farmers and agricultural technicians.

Figure 12. Evidence of prototype heuristic evaluation.

Trustworthiness Matrix

Criterion	Design Evidence	Evaluation Finding	Risk Level	Next Action
Transparency	Confidence score, alternatives, visual evidence, explanation text.	Strong for prototype. Users can see why AI suggested the result.	Medium	Validate explanation clarity with farmers.
Human control	Accept, reject, retake photo, report error, consult technician.	Strong. AI remains assistive and does not execute treatment.	Low	Keep controls visible on every result screen.
Privacy	Consent screen separates required and optional data use.	Good. Needs final privacy wording and retention policy.	Medium	Add deletion request and retention period.
Safety	Low-confidence warning and expert consultation path.	Good. Reduces blind trust and wrong treatment risk.	High	Pilot with real disease/pest cases and review false negatives.

Criterion	Design Evidence	Evaluation Finding	Risk Level	Next Action
Fairness / bias	Data plan includes varied lighting, angles, growth stages, local classes.	Partial. Bias mitigation depends on final dataset quality.	High	Collect diverse local images and monitor per-class errors.
Accessibility	Large text, high contrast, icons, local language, read-aloud option.	Good for low-technical and outdoor users.	Medium	Test readability outdoors and with older farmers.
Reliability	Image quality check, retake-photo flow, audit log.	Good prototype logic. Offline behavior needs implementation.	Medium	Add offline draft saving and failed-upload recovery.
Governance	Prediction log, model version, consent status, feedback outcome.	Good framework. Needs assigned review roles.	Medium	Define reviewer schedule and escalation criteria.

Evaluation Findings

The design meets the core HCAI requirements at the prototype-documentation level. Its strongest areas are decision support alignment, explainability, and human override. The main remaining risks are dataset bias, false-negative disease cases, and the absence of live field testing evidence. These should be addressed before full deployment.

7. Governance and Deployment Plan

Implementation Plan

Phase	Action	Output
1. Prototype finalization	Finalize UI screens, navigation, consent wording, and result controls.	Approved clickable prototype.
2. Dataset preparation	Collect, label, clean, balance, and document rice-plant images.	Dataset card and label guide.
3. Model development	Train CNN, validate with unseen images, calibrate confidence, test top-3 predictions.	Model report and performance metrics.
4. Pilot testing	Test with farmers and technicians using real field scenarios.	Usability findings and error cases.
5. Controlled deployment	Deploy advisory module with monitoring, audit log, and support workflow.	Initial working system.
6. Continuous improvement	Review feedback, correct labels, update model, publish version notes.	Safer and more accurate future versions.

Stakeholders

Stakeholder	Role
Rice farmers	Capture images, review AI output, decide final action, submit feedback.
Agricultural technicians	Validate difficult cases, review reported errors, support farmer education.
Farm cooperatives	Coordinate pilot testing, collect community feedback, promote sustainable use.
System administrators	Maintain user access, data storage, audit logs, and model version records.
AI/data team	Prepare dataset, train/evaluate model, monitor risks, and update recommendations.
Instructor/review panel	Evaluate final deliverable, prototype completeness, and HCAI alignment.

Sustainability

The project supports sustainability by promoting early detection, reducing unnecessary chemical use, encouraging preventive practices, and strengthening farmer access to agricultural information. Long-term sustainability depends on local dataset improvement, technician participation, clear maintenance responsibility, and continuous feedback collection.

Ethical Considerations

The system must not overstate certainty. It should explicitly communicate that the result is advisory, not a guaranteed diagnosis. It must avoid collecting unnecessary personal data, must ask consent before using uploaded images for model improvement, and must provide human review for uncertain or high-risk cases.

Governance and Deployment Process

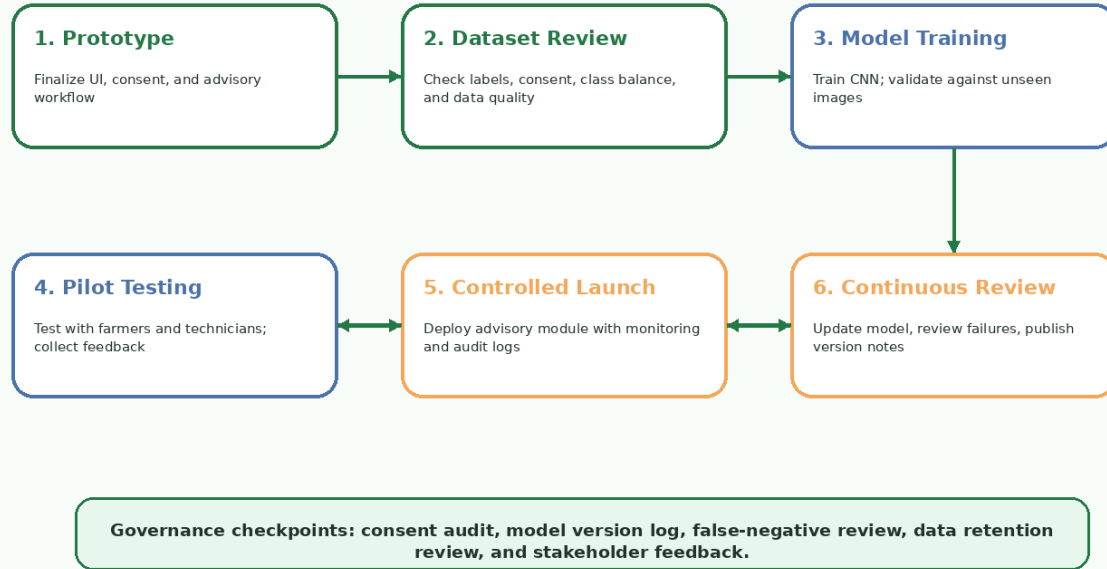


Figure 13. Governance and deployment process diagram.

8. Final Output and Demo

Summary

The final proposed output is a mobile AI advisory module for rice farmers. The system captures a rice-plant image, analyzes it using a CNN-based image classifier, displays an explainable pest/disease advisory result, and allows the farmer to review or reject the output before taking action.

Key Features

Feature	Description
Easy image capture	Farmers take a rice plant photo using a mobile device.
AI diagnosis support	CNN model identifies likely pest, disease, or healthy condition.
Explainable result	Result includes confidence, alternatives, visual evidence, and simple explanation.
Actionable recommendations	The system provides treatment, prevention, and best-practice guidance.
Human override	Farmer can accept, reject, retake, report, or consult technician.
Privacy controls	Consent screen separates required analysis from optional data use.
Accessibility options	Large text, high contrast, icons, read-aloud, and local-language support.
Feedback and audit	User corrections and system logs support future improvement and accountability.

Expected Impact

The expected impact is earlier crop-health awareness, faster farmer response, reduced crop loss, improved productivity, better communication with agricultural technicians, and more responsible use of AI in agriculture. The system strengthens farmer decision-making without removing human judgment.

Demo Flow

Step	Demo Action	What to Explain
1	Open the app and show the welcome screen.	Explain that it is a mobile advisory tool for rice crop-health support.
2	Show consent screen.	Point out required analysis consent and optional data-sharing choices.
3	Open camera guide and capture/review image.	Explain image quality guidance and retake-photo support.
4	Run AI analysis.	Explain preprocessing, CNN classification, confidence, and recommendation mapping.
5	Show diagnosis result.	Highlight confidence, top alternatives, and visual evidence.
6	Show advice and override options.	Emphasize that the farmer remains the final decision-maker.

Step	Demo Action	What to Explain
7	Submit feedback or report error.	Explain feedback loop, model improvement, and governance audit.

Final System Outputs and Prototype Highlights



Expected impact: earlier crop-health awareness, reduced crop loss, improved farmer confidence, and better communication with agricultural support workers.

Figure 14. Final system outputs and prototype highlights.

Appendix: Submission Readiness Checklist

Requirement	Status in this PDF
One single PDF file	Included.
All required sections	Sections 1 to 8 are included.
Embedded screenshots	Concept sketch, workflow, personas, problem map, architecture, UI screens, navigation flow, evaluation evidence, deployment diagram, and final outputs are embedded.
Clickable prototype link	Clickable placeholder included. Replace with actual Figma or deployed system URL before final submission.
At least 5-8 UI screens	Eight UI screens are included in Figure 10.
Trustworthiness Matrix	Included in Section 6.
Governance and deployment plan	Included in Section 7.
Final output and demo	Included in Section 8.

Source Materials Analyzed

1. HCI project documentation: project summary, background, target users, AI solution, human-centered features, benefits, data requirements, and workflow.
2. AI Design Metaphors Activity: AI pipeline, CNN model, assistive autonomy level, risk/failure analysis, transparency/explainability design, and control mechanism design.
3. IT58 Final Project Details: required single-PDF structure, required screenshots, prototype-link requirement, and 100-point rubric.

Recommended action before submission: replace the placeholder prototype URL with your actual Figma or deployed system link and re-export this PDF if your instructor requires the link to be working.